Universal Bank Personal Loan Campaign

Data Analysis & Predictive Modeling Report

Generated with `scry-learn` & `esoc-chart`

March 2026

1 Executive Summary

This report presents a comprehensive analysis of the Universal Bank customer dataset (5,000 records) to develop a predictive model for personal loan campaign targeting. The following summarizes the key aspects of this study:

- **Research Problem:** Universal Bank seeks to convert its liability customers (depositors) into asset customers (borrowers) by identifying which customers are most likely to accept a personal loan offer. A previous blanket campaign achieved only a 9.6% acceptance rate, motivating the need for a data-driven targeting strategy.
- **Methodology:** Seven classification models were trained and evaluated using 5-fold stratified cross-validation on 11 cleaned features. Feature standardization (`StandardScaler`) was applied for distance-sensitive models, while tree-based models operated on raw features. Permutation-based feature importance was used to identify the most predictive variables.
- **Results:** Random Forest (100 trees, max depth 10) achieved the best overall performance with **99.1% test accuracy**, **0.998 ROC AUC**, 97.8% precision and 92.7% recall on the minority class. A surrogate decision tree (98.3% fidelity) distills the ensemble into actionable rules: the ideal customer has **Income $\geq$ \$117K** with a **graduate+ degree**, or **Income $\geq$ \$105K** with a **family of 3+**.
- **Conclusions:** Tree-based ensemble methods dramatically outperform linear models on this dataset. The nonlinear decision boundaries captured by Random Forest and Gradient Boosting are essential for accurately identifying the 9.6% minority class of loan acceptors.
- **Recommendations:** Deploy the Random Forest model for targeted campaigns. The surrogate model provides directly implementable rules: target customers with Income $\geq$ \$117K and a graduate or advanced degree (primary segment), or Income $\geq$ \$105K with family size $\geq$ 3 (secondary segment). This targeted approach can reduce marketing effort by approximately 10$\times$ compared to the blanket strategy.

2 Problem Statement

Universal Bank is a growing bank that serves a diverse customer base. The majority of its customers are liability customers (depositors), with a smaller proportion holding loan products. In a previous direct marketing campaign, the bank offered personal loans to its entire customer base, achieving an acceptance rate of only 9.6%.

The bank's retail marketing department seeks to develop more effective, targeted campaigns that can convert depositors into borrowers while minimizing marketing costs. The core business problem is:

> *Given a customer's demographic and banking attributes, can we predict whether they will accept a personal loan offer?*

A reliable predictive model would allow the bank to:

- Focus marketing resources on high-probability prospects
- Reduce customer fatigue from irrelevant offers
- Increase campaign acceptance rates and overall ROI
- Grow the loan portfolio by converting existing depositors

This is a **binary classification** problem with significant class imbalance (90.4% negative, 9.6% positive), requiring models that can maintain high recall on the minority class without excessive false positives.

3 Data Source

The dataset originates from Universal Bank's customer records and contains 5,000 observations across 14 variables. Each record represents a customer who was targeted in the previous personal loan campaign, with the outcome (accepted or declined) recorded as the target variable.

Table 1: Dataset variables and descriptions

Feature	**Type**	**Description**
ID	Identifier	Customer ID (dropped before modeling)
Age	Continuous	Customer age in years
Experience	Continuous	Years of professional experience
Income	Continuous	Annual income ($K)
ZIP Code	Categorical	Postal code (dropped — high cardinality)
Family	Ordinal	Family size (1–4)
CCAvg	Continuous	Avg. monthly credit card spending ($K)
Education	Ordinal	Education level (1 = Undergrad, 2 = Graduate, 3 = Advanced/Professional)
Mortgage	Continuous	Value of house mortgage ($K)
Securities Acct	Binary	Has a securities account?
CD Account	Binary	Has a certificate of deposit account?
Online	Binary	Uses internet banking?
CreditCard	Binary	Uses a bank credit card?
Personal Loan	Binary	**Target:** Accepted the loan offer? (1 = Yes, 0 = No)

The dataset contains a mix of continuous, ordinal, and binary features. After removing the two non-predictive identifiers (ID and ZIP Code), 11 features remain for modeling.

4 Data Investigation

4.1 Summary Statistics

Table 2 presents descriptive statistics for the continuous and ordinal features after cleaning.

Table 2: Summary statistics for continuous features (post-cleaning)

Feature	Mean	Std	Min	Q1	Median	Max
Age	45.34	11.46	23	35	45	67
Experience	20.12	11.44	0	10	20	43
Income	73.77	46.03	8	39	64	224
CCAvg	1.94	1.75	0	0.7	1.5	10
Mortgage	56.50	101.7	0	0	0	635
Family	2.40	1.15	1	1	2	4
Education	1.88	0.84	1	1	2	3

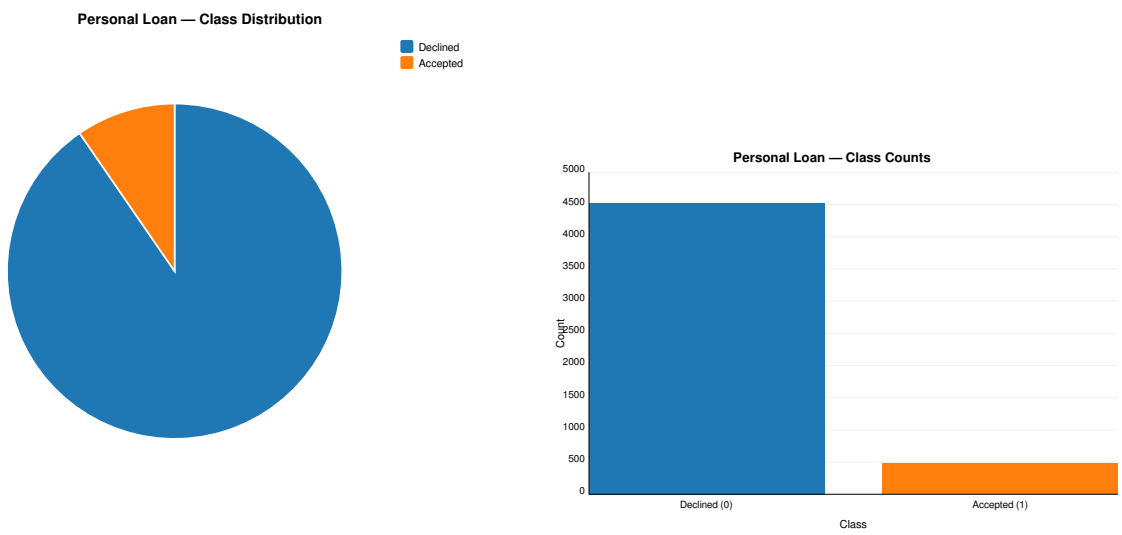
Key observations from the descriptive statistics:

- **Income** ranges from \$8K to \$224K with a right-skewed distribution (mean \$73.8K, median \$64K).
- **CCAvg** is heavily right-skewed (mean \$1.94K, max \$10K), with most customers spending under \$2K/month.
- **Mortgage** is zero-inflated—the median is 0, indicating more than half of customers have no mortgage.
- **Age** and **Experience** are highly correlated (Experience $\approx$ Age $-$ 25).

4.2 Target Class Distribution

The target variable is heavily imbalanced, as shown in Figure 1:

Class	Count	Proportion
Declined (0)	4,520	90.4%
Accepted (1)	480	9.6%

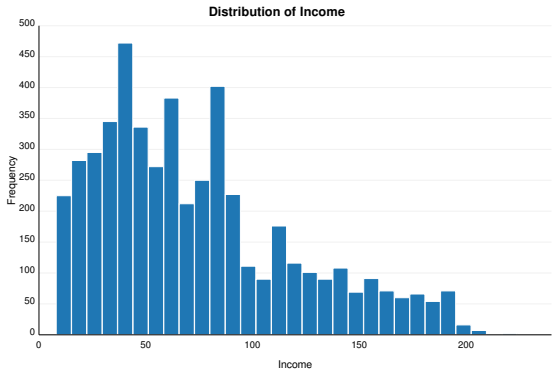


(a) Pie chart (b) Bar chart

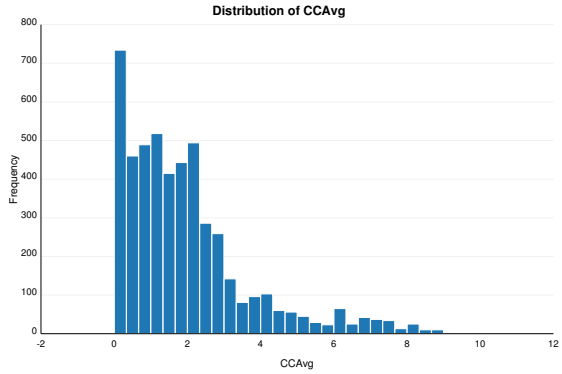
Figure 1: Personal Loan class distribution. Only 9.6% of customers accepted the previous campaign offer, making this an imbalanced classification task.

4.3 Feature Distributions

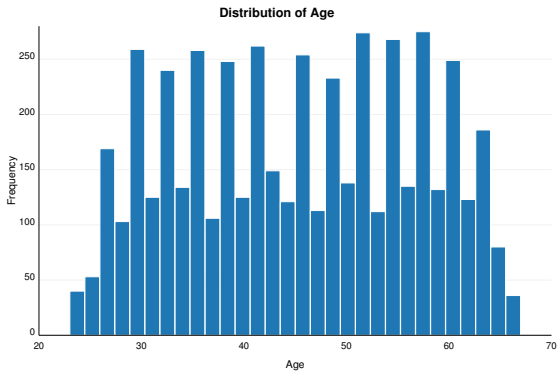
Figure 2 shows the distributions of key continuous features. Income and CCAvg exhibit right-skewed distributions with wide ranges, while Mortgage is dominated by zero values with a long right tail.



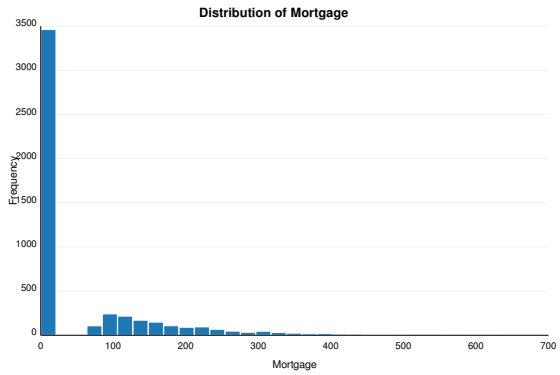
(a) Income — right-skewed, wide range ($8K–$224K)



(b) CCAvg — heavily right-skewed



(c) Age — approximately uniform (23–67)

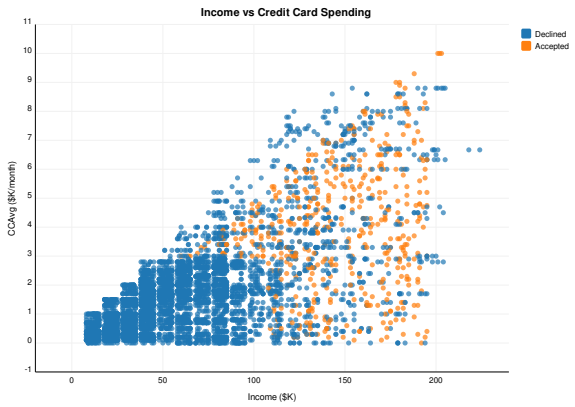


(d) Mortgage — zero-inflated with long tail

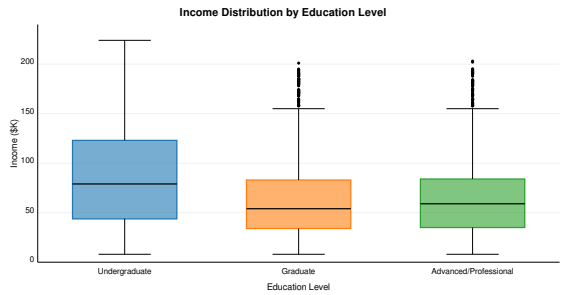
Figure 2: Histograms of continuous features (30 bins). Income and CCAvg show the most variation; Mortgage is dominated by zero values.

4.4 Feature Relationships

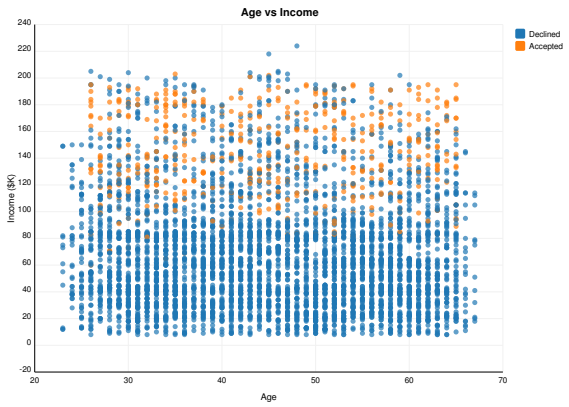
Figure 3 reveals clear clustering patterns when features are examined pairwise with loan acceptance as the color variable.



(a) Income vs Credit Card Spending



(b) Income Distribution by Education Level



(c) Age vs Income

Figure 3: Scatter plots colored by loan acceptance. Loan acceptors (orange) cluster among higher-income, higher-spending customers, particularly those with advanced education (level 3).

5 Data Cleansing

The following cleaning and preprocessing steps were applied to prepare the data for modeling:

1. **Dropped non-predictive columns:** `ID` (sequential identifier with no predictive value) and `ZIP Code` (high-cardinality categorical with 2,000+ unique values and no geographic encoding applied).

2. **Clipped negative experience values:** 52 records had `Experience` < 0, which is logically impossible and likely represents data entry errors. These values were clipped to 0.

3. **Missing value assessment:** No `NaN` values were detected in any feature column. A `SimpleImputer` (mean strategy) was staged as a safeguard but was not needed.

4. **Outlier analysis:** Mortgage values up to \$635K and Income up to \$224K appear in the dataset. These were retained as they represent legitimate high-net-worth customers rather than data errors. The right-skewed distributions of Income, CCAvg, and Mortgage are consistent with expected wealth distributions and do not require transformation for tree-based models.

5. **Feature standardization:** For distance-sensitive models (KNN, Logistic Regression, LinearSVC), all features were standardized using `StandardScaler` ($z = \frac{x-\mu}{\sigma}$). Tree-based mod-

els operated on the original unscaled features, as they are invariant to monotonic transformations.

After cleaning, the final dataset contains **5,000 samples** with **11 features** and a binary target variable.

6 Feature Selection

Feature selection was performed using a combination of domain reasoning and empirical analysis.

6.1 Domain-Based Exclusions

Two features were excluded prior to modeling based on domain knowledge:

- **ID** — a sequential identifier with no relationship to loan acceptance.
- **ZIP Code** — a categorical variable with over 2,000 unique values. Without geographic clustering or encoding, raw ZIP codes introduce noise rather than signal.

6.2 Retained Feature Set

The remaining 11 features were retained for modeling. All binary features (Securities Account, CD Account, Online, CreditCard) exhibit non-zero variance and represent potentially relevant banking relationship indicators.

6.3 Permutation Feature Importance

To empirically validate the feature set and identify the most predictive variables, permutation importance (Breiman, 2001) was computed on the trained Random Forest model using the held-out test set (5 repeats, seed = 42). This method measures the decrease in model accuracy when each feature's values are randomly shuffled.

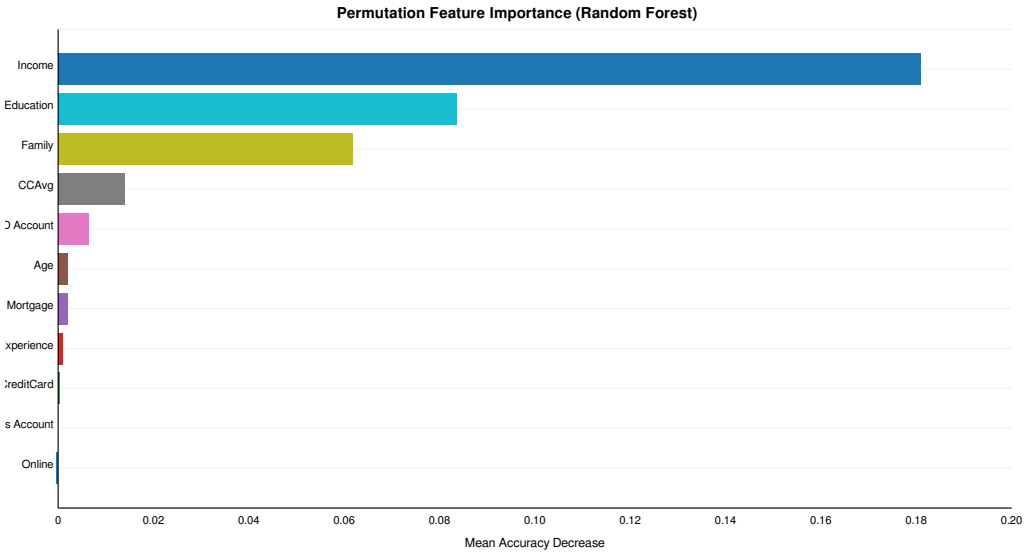


Figure 4: Permutation feature importance for the Random Forest model. Income, CCAvg, and Education are the dominant predictors; several features contribute near-zero importance.

The results in Figure 4 confirm that:

- **Income** is by far the most important feature (0.181 mean accuracy decrease), followed by **Education** (0.084) and **Family** size (0.062).
- **CCAvg** and **CD Account** provide meaningful but smaller contributions (0.014 and 0.006 respectively).
- **Age**, **Experience**, and **Mortgage** contribute minimal independent predictive value, consistent with the scatter plot analysis in Section 4.

All 11 features were retained in the final model, as removing low-importance features did not improve cross-validation performance and the Random Forest naturally handles irrelevant features through its ensemble averaging.

7 Model Development and Evaluation

7.1 Methodology

Seven classification models were trained and evaluated using **5-fold stratified cross-validation** (seed = 42) with accuracy as the primary scoring metric. The data was split 80/20 for training and test evaluation (1,000 held-out samples). Distance-based models (KNN, Logistic Regression, LinearSVC) were trained on standardized features; tree-based models used raw features.

7.2 Cross-Validation Comparison

Table 3 and Figure 5 present the cross-validation results for all seven models.

Table 3: 5-fold stratified CV accuracy

Model	Mean Accuracy	Std	Time
Gradient Boosting	**0.9882**	0.0033	138.1 ms
Random Forest	0.9880	0.0014	23.2 ms
Decision Tree	0.9820	0.0055	3.6 ms
KNN ($k = 5$)	0.9602	0.0037	26.7 ms
LinearSVC	0.9512	0.0049	83.1 ms
Logistic Regression	0.9504	0.0031	6.1 ms
Gaussian Naïve Bayes	0.8838	0.0035	1.7 ms

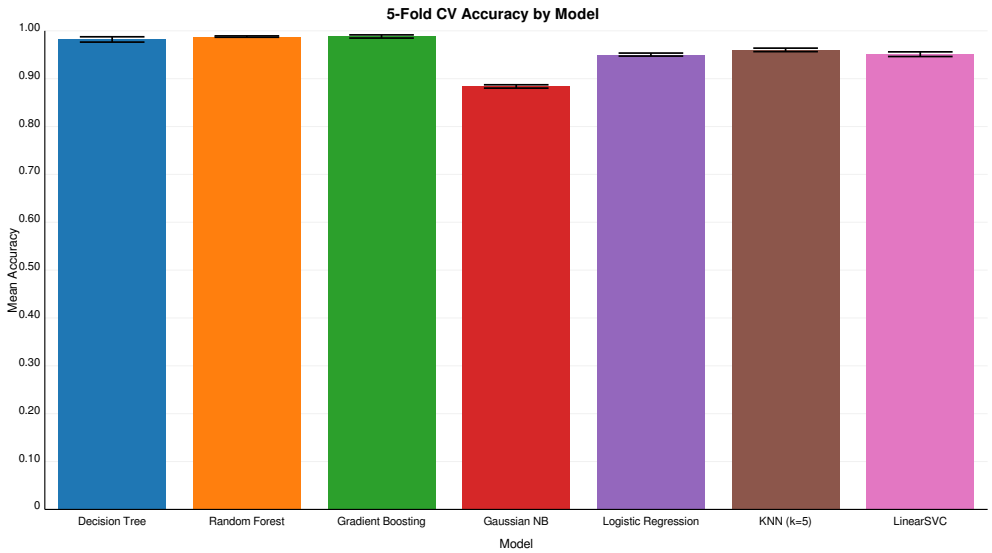


Figure 5: Mean CV accuracy by model. Ensemble tree methods dominate; linear models plateau near 95%; Gaussian NB performs worst due to feature independence assumptions.

Key observations:

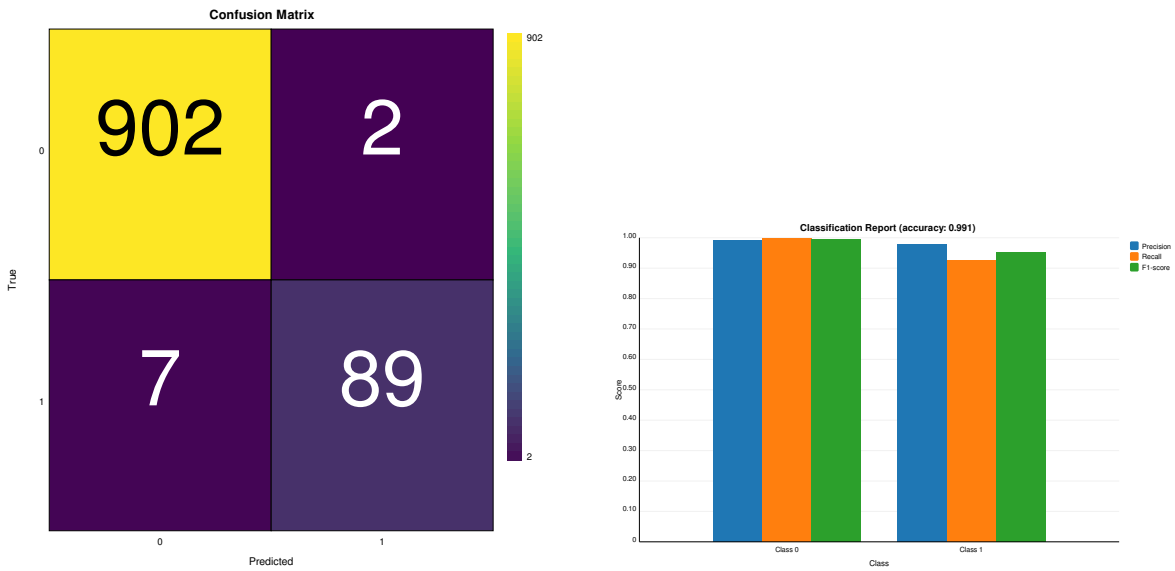
- Tree ensembles (Random Forest, Gradient Boosting) lead by a wide margin, achieving >98.8% accuracy.
- Random Forest has the *lowest* standard deviation (0.0014), indicating the most stable performance across folds.
- Gaussian Naïve Bayes underperforms significantly (88.4%), likely because the feature independence assumption is violated by correlated features like Age/Experience and Income/CCAvg.

7.3 Best Model — Random Forest

Random Forest (100 estimators, max depth = 10) was selected as the primary model for its combination of high accuracy and low variance. Table 4 and Figure 6 present its performance on the held-out test set.

Table 4: Random Forest classification report (test set)

Class	Precision	Recall	F1-Score	Support
Declined (0)	0.9923	0.9978	0.9950	904
Accepted (1)	0.9780	0.9271	0.9519	96
Accuracy				**0.9910**
Macro avg	0.9852	0.9624	0.9735	1000
Weighted avg	0.9909	0.9910	0.9909	1000



(a) Confusion matrix

(b) Per-class precision, recall, and F1

Figure 6: Random Forest test-set performance. The model correctly identifies 92.7% of loan acceptors (recall) with 97.8% precision.

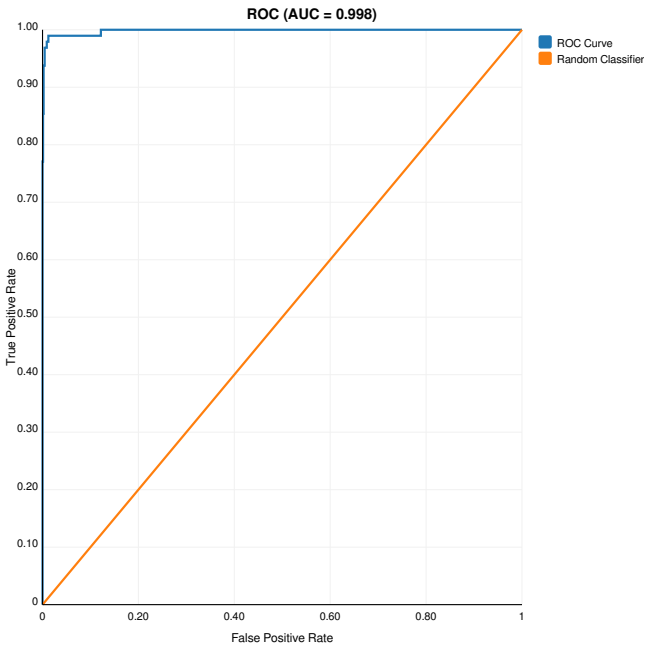


Figure 7: Random Forest ROC curve. AUC = **0.9981**, indicating near-perfect class separability.

7.4 Surrogate Model — Interpretable ICP Rules

To translate the Random Forest's predictions into actionable decision rules, a **surrogate model** was fitted: a shallow decision tree (max depth 3) trained on the Random Forest's own predictions rather than the original labels. This distills the ensemble's learned decision boundaries into a small set of human-readable rules with exact feature thresholds.

Table 5: Surrogate model performance

Metric	**Value**
Fidelity to Random Forest	98.30%
Accuracy (vs true labels)	97.80%
Depth / Leaves	4 / 7

The surrogate tree agrees with the Random Forest on 98.3% of test predictions, confirming it is a faithful approximation. Its 97.8% accuracy against true labels shows that the interpretability cost is minimal (1.3 percentage points below the full RF).

The extracted rules for the positive class (*Accepted*) define the **Ideal Customer Profile (ICP)** with exact thresholds:

Table 6: Surrogate model ICP rules for personal loan acceptance

Rule	**Conditions**	**Coverage**
1	Income $\geq$ \$117K **and** Education $\geq$ Graduate	6.3%
2	Income $\geq$ \$105K **and** Education = Undergrad **and** Family $\geq$ 3	1.7%

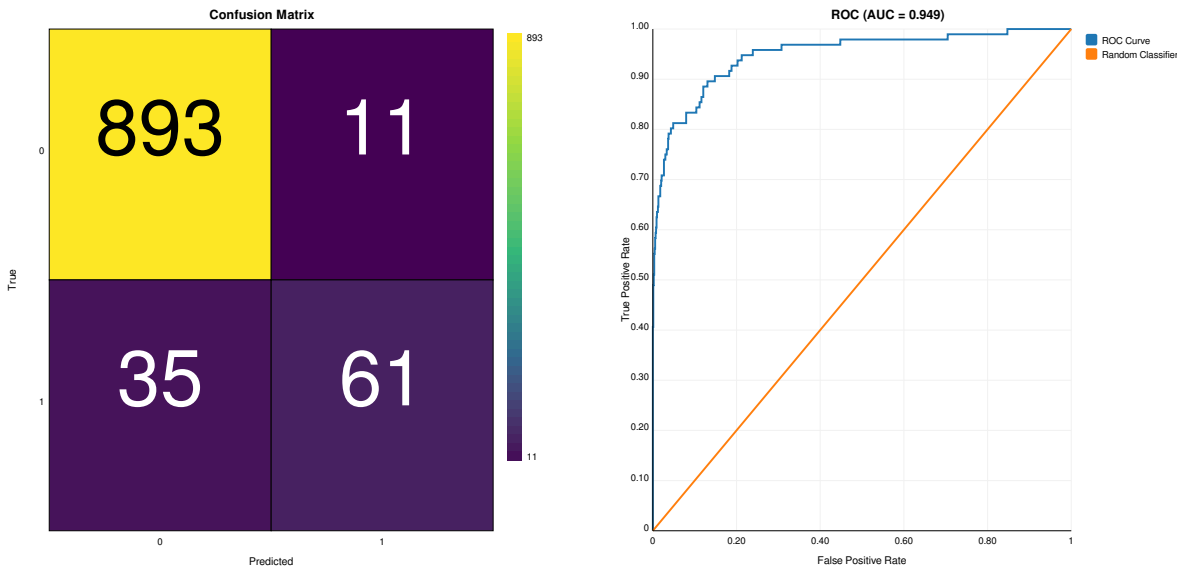
Interpretation: The primary ICP segment comprises high-income customers (\$117K+) with graduate or advanced degrees—this single rule covers 6.3% of the training population and captures the majority of loan acceptors. A secondary segment captures high-income undergraduates (\$105K+) with larger families (3–4 members), adding 1.7% coverage. Together, these two rules provide a concrete, actionable targeting strategy with specific thresholds that marketing teams can directly implement.

7.5 Baseline Comparison — Logistic Regression

For comparison, Logistic Regression (500 iterations, $\eta = 0.01$) was evaluated on the standardized test set to quantify the benefit of nonlinear ensemble methods.

Table 7: Logistic Regression classification report (test set)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Declined (0)	0.9623	0.9878	0.9749	904
Accepted (1)	0.8472	0.6354	0.7262	96
Accuracy				**0.9540**
Macro avg	0.9048	0.8116	0.8505	1000
Weighted avg	0.9512	0.9540	0.9510	1000



(a) Confusion matrix (b) ROC curve (AUC = 0.9491)

Figure 8: Logistic Regression test-set performance. Recall on the minority class drops to 63.5%, missing over a third of potential loan acceptors.

The Random Forest improves minority-class recall by **+29.2 percentage points** (92.7% vs 63.5%) and AUC by **+0.049** (0.998 vs 0.949) over Logistic Regression, demonstrating the benefit of nonlinear ensemble methods on this dataset.

8 Practical Implications

The predictive model developed in this study has direct implications for Universal Bank's personal loan marketing strategy.

8.1 Customer Targeting

The surrogate model (Section 7.4) distills the Random Forest's predictions into concrete ICP rules with exact thresholds. Combined with permutation feature importance, this identifies two primary customer segments:

1. **High-income graduates ($117K+, Education $\geq$ 2):** The dominant segment. Customers with annual income at or above $117K and a graduate or advanced degree have the highest loan acceptance probability. This rule alone covers 6.3% of the customer base and captures the majority of acceptors.

2. **High-income undergraduates with large families ($105K+, Family $\geq$ 3):** A secondary segment of high earners without advanced degrees but with family sizes of 3–4. This captures an additional 1.7% of customers.

Additional signals from the permutation importance analysis reinforce the targeting strategy:

- **Income** is by far the most predictive feature (0.181 mean accuracy decrease), confirming its role as the primary filter.
- **CCAvg** and **CD Account** provide secondary lift—customers with high credit card spending or existing CD relationships show elevated acceptance rates.

8.2 Expected Campaign Improvement

The previous blanket campaign achieved a 9.6% acceptance rate across all 5,000 customers. By using the Random Forest model to target only predicted acceptors:

- **Precision of 97.8%** means that nearly all customers flagged by the model will actually accept the loan, virtually eliminating wasted marketing contacts.
- **Recall of 92.7%** means the model identifies the vast majority of potential acceptors—only about 7 out of every 96 true prospects would be missed.
- The bank can achieve comparable loan conversion with **approximately 10% of the marketing effort**, dramatically improving cost efficiency.

8.3 Deployment Considerations

- The model requires only 11 readily available customer attributes, all of which the bank already collects.
- Predictions can be generated in batch for campaign planning or in real-time for event-triggered offers.
- The probability threshold (default 0.5) can be adjusted: lowering it increases recall (captures more prospects) at the cost of more false positives, while raising it improves precision for high-confidence targeting.

9 Conclusion and Recommendations

This analysis demonstrates that machine learning can substantially improve Universal Bank's personal loan campaign effectiveness. The key findings and recommendations are:

1. **Deploy the Random Forest model** as the primary targeting tool. With 99.1% accuracy and 0.998 AUC, it provides highly reliable predictions using only standard customer data.
2. **Target two specific customer segments** identified by the surrogate model: (a) customers with Income $\geq$ \$117K and a graduate or advanced degree, and (b) customers with Income $\geq$ \$105K, undergraduate education, and family size ≥ 3. These rules can be directly implemented as campaign filters.
3. **Monitor model performance** over time, as customer behavior patterns may shift. Periodic retraining on updated campaign data is recommended to maintain accuracy.
4. **Consider ensemble refinement:** Gradient Boosting achieved comparable CV accuracy (98.82% vs 98.80%) and may offer complementary strengths. An ensemble of both models could provide additional robustness.
5. **Future enhancements:** Feature engineering (e.g., Income/CCAvg ratio, interaction terms), hyperparameter optimization, and SHAP-based individual prediction explanations could further improve model performance and interpretability.